Thesis Report
A Hybrid Model Including Bi-RNN,Attention Layer and Lexicon-Based Feature Extraction For Detecting Frustration On Online Product Reviews

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DECLARATION

This thesis proposal is submitted to the Department of Information and Communication Engineering (ICE), Noakhali Science and Technology University, Noakhali -3814, in partial fulfillment of the requirments for having the B.Sc. degree in ICE under the course entitled with “ICE – 4218”. So, I, here by declare that this thesis report has not been submitted elsewhere for the requirment of any kind of degree, diploma or publication.

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ACCEPTANCE

This Thesis proposal is submitted to the Department of Information and Communication Engineering (ICE), Noakhali Science and Technology University, Noakhali -3814, in partial fulfillment of the requirments for having the B.Sc. degree in ICE under the course entitled “ICE - 4218”.

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To begin, I want to express my thanks to the almighty creator for providing me with the strength and golden chance to conduct my thesis proposal work — "A HYBRID MODEL INCLUDING BI-RNN,ATTENTION LAYER AND LEXICON-BASED FEATURE EXTRACTION FOR DETECTING FRUSTRATION ON ONLINE PRODUCT REVIEWS". I also grateful and would like to express my sincere gratitude to my supervisor, Md. Sabbir Ejaz sir for his germinal ideas, invaluable guidance, continuous encouragement, and constant support in making this research possible.Without his assistance this thesis proposal paper would be extremely diffcult for me to complete. I am quite appreciative of all of his assistance Additionally, I would like to express my gratitude to all of my teachers at the Department of Information and Communication Engineering (ICE) for their insightful suggestions and academic assistance during the course work. I am also grateful to all of my friends who have supported me in my endeavor. Finally, I acknowledge my sincere indebtedness and gratitude to my parents for their love, support, dream and sacrifice throughout my life.

ABSTRACT

The capability to identify the emotional status of an individual based on online text has grown in value over the past few years in application areas like mental management, customer care and character analysis. In this paper a hybrid deep learning-based framework to automatically recognize frustration in online reviews using both semantic and affective information is proposed. In the information in dataset, user-generated reviews forms the input; it is preprocessed with the NLTK based approaches that incorporates tokenization and lemmatization tags, part-of-speech tagging as well as a removal of stop-words, so as to provide a clean textual input. In order to deal with the problem of class imbalance and data diversity, data augmentation of minority (frustrated) class using NLPAug library is performed with synonyms. Emotion-based features are inferred with NRCLex, which computes the scores of frustration-related emotions (anger, sadness, fear), whereas sentiment polarity and subjectivity are inferred with TextBlob. This inventions of features based on lexicon is subsequently incorporated as secondary number inputs. GloVe pre-trained embeddings and a Bidirectional SimpleRNN are used to extract pre-trained semantic content of the text and learn expressive temporal behaviour respectively. It uses a bespoke attention mechanism that gives the model attention to emotionally salient tokens in a review. The architectural model uses the concerted effort of both the Bi-RNN and the emotion- based features with the attention-enhanced textual representation into a dense neural classifier. The training includes class weighting, dropout regularization, learning rate scheduling, and early stopping, which are all meant to fend off overfitting. The test set achieves good results when evaluating the model, delivering a high accuracy level and good metrics in detecting frustration. This paper presents an efficient way of detecting frustration by integrating deep contextual knowledge and human-labeled emotional attributes. The method may be generalizable to other hypertensive conditions, which can have implications on using the method in intelligent forms of support systems and real-time emotional checks.

Keyword: Frustration Detection, Sentiment Analysis, NRCLex, TextBlob, Attention Mechanism, GloVe Embedding, Data Augmentation, Emotion Mining, Deep Learning, Online Reviews.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

As the number of user-made contents on digital media continues to vastly expand, the need to interpret human emotions based on the words they use has become ever-more important. Particularly, online reviewing is the wealthiest source of subjective statements, which can bear precious information of emotional condition of a user. Among other emotional states, frustration, in particular, is the one that needs to be pointed out, as it can reflect either dissatisfaction, the problem with the usability of the system, or even the mental distress. Timely recognition of these emotions will have the ability to improve customer support systems, design products, and analyze the real-time experience of its users.

Common sentiment analysis methods are a representation of a given textual data through broad categories like positive, negative, or even neutral. Nevertheless, these categories tend to ignore more subtle states of emotion, such as frustration, which co-exist alongside a mixture of feelings or are intertwined in more subtle phrases. Consequently, additional more advanced methodologies mixing lexical and discursive comprehension are essential in effective frustration identification.

Our proposal is in such a research, a hybrid deep learning solution that incorporates semantic and affective features to determine the occurrence of frustration in online textual reviews. This architecture of the model takes into consideration a Bidirectional Recurrent Neural Network (Bi- RNN) combined with a self-attention mechanism to improve the contextual relationships and focus on emotionally meaningful words. We also retrieve lexicon-based affective features like frustration-related emotion scores (e.g. anger, sadness, fear) with NRCLex lexicon, sentiment polarity and subjectivity with TextBlob. This aspect is used together with the deep semantic titling in order to make the model sensitive to emotions.

Data augmentation is done by replacing words with synonyms to overcome the class imbalance problem that is prevalent in emotional dataset some of which will only contain neutral or non- frustrated reviews. In addition, pre-trained GloVe embeddings are applied to give rich semantic representation of words and this contributes to the ability of the model to generalize more on unseen data.

The system is tested on an actual data of online reviews. Standard performance results are determined by using accuracy, precision, recall, F1-score, and confusion matrices. Findings indicate that incorporation of attention mechanisms and affective lexicon features shows great importance in enhancing the capability of the model to detect frustration, more availing than the traditional binary sentiment models.

The research is relevant to the emerging area of affective computing and emotion-sensitive artificial intelligence which has been used in such areas as customer support automation and other fields.

1.2 Background Study

The rise of online communication has generated vast amounts of textual data, such as reviews and feedback, which are valuable for understanding customer satisfaction and emotional well- being.Traditional sentiment analysis mainly focuses on classifying text as positive, negative, or neutral, but recent

research emphasizes the need to detect fine-grained emotions like frustration—often a sign of unmet expectations or negative experiences. Frustration is subtle and often overlaps with emotions like anger or sadness, making it difficult to capture using basic polarity-based tools like TextBlob or VADER.

To address this, researchers have begun combining affective lexicons (e.g., NRCLex, EmoLex) with machine learning models, although lexicon-only methods often fail to grasp context and semantics. Deep learning techniques, especially Bidirectional RNNs and attention mechanisms, have shown strong performance in emotion detection by better capturing linguistic dependencies and focusing on emotionally charged phrases. Challenges like class imbalance are addressed using data augmentation and class weighting.

This study proposes a hybrid model combining NRCLex emotion features, TextBlob sentiment scores, Bidirectional SimpleRNN, and an attention mechanism, along with GloVe embeddings. This multidimensional approach enhances frustration detection by bridging symbolic and contextual emotion understanding, offering a robust solution for affective computing and textual emotion analysis.

1.3 Problem Statement

Online reviews become an essential tool of feedback nowadays not only in the domain of e- commerce platforms and services but also in applications. User sentiments are usually present in such reviews, and these sentiments could be either satisfaction or dissatisfaction. However, more advanced sentiment analysis approaches may be required to more precisely recognize the underlying emotion in a particular piece of text, more than just a general positive or negative sentiment time and time again failing to discern more specific and nuanced emotional state such as frustration that can be an early indicator of user churn, dissatisfaction, or service failure.

Frustration is not just a plain negative emotion but a complex set of emotions that can be caused due to a lack of expectations, a failure, or hurdles and is a rather common phenomenon. Frustration is especially difficult to identify correctly in written reviews, owing to the delicacy and subjectivity of writing, and to the sparseness of labeled frustration data. In addition, current methods used are mostly either using lexicon based features or using classical machine learning approaches and these methods are typically not deep enough or flexible enough to capture frustration.

The proposed research will aim at narrowing the existence of this gap with the design and development of a hybrid deep learning based that will combine lexicon-based features with contextual neural networks. The model outlined aims to improve on accuracy in the detection of frustration by uniting semantics embedding, recurrent architecture and attention segmentations.

1.4 Motivation

Over the past years, online product reviews have turned out to be a primary influential source of customer and company feedback. Nonetheless, as part of a simple positive/negative sentiment, there is also frustration associated with many reviews, which is a highly specific and very negative emotional indicator that reflects anger or disappointment and a sense that something is not right, expectations have not been met, or the user experience is not good. The nicer borderline emotion of frustration gets lost in typical sentiment analysis models that tend to bucket it out based upon the negative words being present.

Hence, the frustration can only be recognized in a more context-sensitive and subtle way. We use a Bidirectional Recurrent Neural Network (Bi-RNN) to overcome this problem because the latter can extract the temporal information of textual data in both directions thus letting the model comprehend the entire context of a review. Moreover, lexicon-based features (e.g. NRCLex emotion scores, polarity, subjectivity) are added to introduce expert knowledge into the model in the area of investigation. The hand-made linguistic characteristics support the model in discovering frustration-expressions which may be sparse or dormant in the training materials. Lastly, an attention layer is added to automatically grow concentration on the information-rich areas of the review, so that the model could attach more prominence to the particular words or phrases that reveal frustration. It is a hybrid architecture that takes the best of both worlds deep contextual representation and human interpretable feature representation and leads to a more accurate and robust frustration detection system.

1.5 Objectives

The main target of this study is to develop smart system, which can sense frustration through online reviews on the basis of deep learning. The following is the list of specific objectives:

- To tokenize, lemmatise and part-of-speech tag the textual data to gain more linguistic understanding out of it.
- To obtain lexicon based emotional terms like frustrations score, sentiment polarity, subjectivity and emotional diversity with the help of NRCLex and TextBlob.
- As word augmentation techniques such as synonyms can be used to augment the dataset to balance the class distribution in preparation for a better model generalisation.
- To build a hybrid deep learning architecture, a Bidirectional Simple RNN with an attention layer, to learn features both sequentially as well as contextually salient.
- To combine semantic word embeddings (GloVe) with the affective features to increase the contextualization of the model to the affective interpretation.
- To test the model using accuracy, classification report, and confusion matrix on the real world online review datasets.
- In order to show the capability of attention-enhanced RNN models to capture frustration when compared to the conventional sentiment classification models.

1.6 Thesis Outline

There are five chapters in this thesis. Chapter 1 (Introduction) gives us a summary on the research topic highlighting the need to detect the frustration in online review, the motivation of the study and the scope of the study. The second chapter includes Literature Review (Chapter 2), which is a summary of the available literature on sentiment analysis, emotion detection, lexicon-based learning, and deep learning approaches on textual emotion classification. In Chapter 3 (Methodology) reports the whole procedure used in the study, including preprocessing of the data, feature extraction, architecture of model and training process. Chapter 4 (Result Analysis) contains the discussion of performance of the applied model on different evaluation metrics, discusses the results, and compares them with baseline methods.

Lastly, Chapter 5 (Conclusion) points out the highlights of the major findings, describes the limitations of the current works and suggests directions and idea to future developments and hopes.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Consumer sentiments, opinion, and emotional state can be measured because of the many online reviews that are made on the online platforms as a result of the popularity of user-generated content. Out of a pool of emotions, one of the most important sentiment that demonstrates a state of dissatisfaction among users, is the indicator of frustration that is related to poor user experience, unmet expectations or limitations posed by the system. There has been a much interest in detecting such emotions, in particular frustration, as well in natural language processing (NLP) and affective computing. The traditional approaches of sentiment analysis often label the text materials as negative, positive or neutral, and thus fail to capture more subtle moods such as frustration under the negative label. Thus, more detailed standpoints about the emotional points of view were approached by the researchers to use emotion-specific classification technologies.

This chapter gives a review of the previous studies that have been done that are concerned with emotion and frustration identification of textual data. It discusses three key topics: (i) sentiment and emotion analysis, with an emphasis in how to extract emotional states in text; (ii) lexicon based technique, such as; NRCLex, TextBlob, which provides rule-based emotional cues; and (iii) deep learning approach, including; recurrence neural networks (RNN), bi-directional model and attention mechanism, which is more context and semantically rich information in emotion classification. The chapter also looks at recent works which combine the features of lexicon and neural networks to enhance emotional comprehension. The review forms a premise of knowledge and figures out essential research gaps that the frustration detection system that relates to the proposed deep learning-based system seeks to bridge.

2.2 Related Works

The article by Suri S et ai.[1] is concerned with the analysis of negative reviews based on the Amazon fine food review data. Word clouds were applied in finding keywords relating to buyer frustration. we used Logistic Regression and obtained 86.96% accuracy which is the most accurate compared to Naive Bayes classifier whose value of accuracy was 85.7. The non parametric classification technique of KNN gave an accuracy of 66%. Compared to Naive Bayes, Logistic Regression showed more improvement of 1.26 percent. The study will increase the knowledge of customer dissatisfaction by use of powerful data analyses.

Sallam A et ai.[2] studies how to detect emotion in words with complicated machine learning and deep learning frameworks. It can be applied to customer feedback, psychological research and media analysis. The differing linguistic, cultural and contextual issues are the challenges. The paper used Logistic Regression, Extra Trees, and SGD, as well as the deep learning models of BiLSTM, BERT, and RoBERTa with feature extraction methods of TF-IDF, N- grams, Word2Vec, and FastText. The precise performance was found to be 93.67 and BERT performed best in regards to accuracy using datasets ISEAR and AIT-2018.

The analyses of techniques of sentiment analysis with Twitter data are given in Khan M et ai.[3]. Classical (e.g. Logistic Regression, SVM), state of the art (e.g. RNNs, CNNs) and hybrid approaches

between traditional and deep learning approaches are addressed. Some of the challenges identified in the research include noisy information and contextual fluctuation. It is focused on hybrid models as a potential way of enhancing sentiment analysis.

Chutia T et ai.[4] considered 330 articles (2013-2023) which dwell on deep learning methods applied in recognition of human emotions. The research demonstrated how effective models such as BERT, BiLSTM are in the analysis of text with shades of meaning. Open issues include multilingual dataset support as well as dynamic emotion detection. The possible solutions include combining multimodal data and using pre-trained models to ensure improved precision and flexibility.

Wang X et ai. [5] surveys the neural network architectures such as CNNs, RNNs, and ransformers to perform sentiment analysis. Making use of such datasets as product reviews and social media and paying attention to LSTM and BERT, the research highlights their better functioning in handling long text dependencies and applying transfer learning to small data volumes. The future work will be to make models more interpretable and integrating multimodal data.

Nandwani P et al. In [6], the author compares lexicon-based, machine learning and deep learning approaches to sentiment and emotion analysis. Lexicon-based methods are independent of labeled data; rather predetermined dictionaries are used. Machine learning applications are dependent on labeled dataset. Such deep learning solutions as RNNs and LSTMs are better at working with long-term text dependencies. Transfer learning is known to receive the models towards the new domains where there exists limited labeled data. The analysis of such datasets like SemEval, SST, and ISEAR demonstrated better results of deep learning, especially on large data.

As Patel P et ai.[7] in their article, Emotion Detection in Text: A Deep Learning Approach to Sentiment Analysis, notice, this possibility is rather important to discuss due to the limited efficiency of the previous approaches to the sentiment analysis issue. It has developed a carefully marked dataset to label emotion like happy, anger and sadness. BERT showed the possibility to interpret complex patterns of language with an accuracy of 92 percent in performing sentiment analysis and 90 percent in performing emotion recognition (BERT). The study will help in NLP and in enhancing sentiment analysis, interpretation of customer feedback and applications in mental health.

In the article entitled SentiLSTM: A Deep Learning Approach to Sentiment Analysis of Restaurant Reviews written by Eftekhar Hossain et al.[8]; a novel approach identifying sentiment has been presented that focuses on the Bengali restaurant reviews. The authors consider the issues concerning the lack of availability of annotated datasets and the impossibility to capture sentiment recognition accurately due to the low-resource languages such as Bengali.The authors explore the manual annotated corpus that consists of 8,435 Bengali restaurant reviews that are either positive or negative. The accuracy of 91.35% achieved by the model is impressive and surpasses classical machine learning-based models.This research identifies the need of a automated sentiment detection to make better decisions and to carry out better services by the one providing the services and by the customer of the restaurant.The authors mention the necessity of expanding the dataset and to include more sentiment classes to further enhancing the model. This work can contribute in an effective manner to the domain of natural language processing due to exhibiting a successful deep-learning based representation of Bengali language sentiment detection. The results of its work highlight the possibility of such models in restaurant industry and other settings.Automated sentiment detection to make better decisions and to

carry out better services by the one providing the services and by the customer of the restaurant.The authors mention the necessity of expanding the dataset and to include more sentiment classes to further enhancing the model.This work can contribute in an effective manner to the domain of natural language processing due to exhibiting a successful deep-learning based representation of Bengali language sentiment detection. The results of its work highlight the possibility of such models in restaurant industry and other settings.

An article by Amjad Iqbal et al. [9](2022) entitled Sentiment analysis of consumer reviews using deep learning discusses the shortcomings of sentiment analysis (SA) of consumer reviews and how deep learning methods can be utilized to address them; more particularly, how such approaches can be used in analyzing text polarity and Deriving insights about unstructured feedback. The framework that the authors suggest involves multiple processes of SA that include data collection, preprocessing (tokenization, removal of punctuation, removal of hashtags), feature encoding based on word embeddings, and three LSTM-based models with different architecture uses to perform classification. They compare their method to various datasets, such as Amazon ones, IMDB, and Yelp, stating that Model 1, a single LSTM layer with embedding and dense layers, had better accuracy rates than other models, as well as the conventional models such as SVM and CNN. The crucial importance of comprehensive preprocessing, versatile feature encoding and strong model formulation to obtain accurate results of sentiment classification is highlighted in the course of the study and indicates the promising future of LSTM-based models in the analysis of consumer feedback.

The article by Soundearajah et ai.[10] (2022) discovers the application of machine learning to conduct sentiment analysis on the ASOS product reviews. The study revolves around the study based on reviews on over 3, 000 reviews and they are categorized under positive (excellent), neutral, (average) and negative (bad). The authors used the state-of-the-art preprocessing and feature extraction methods and designed different machine learning models, the Nu-Support Vector Classification model making the most accurate prediction (85.99%). The paper attributes the usefulness of sentiment analysis in the identification of operational challenges and improvement of customer satisfaction by indicating that delivery-related complaints implied logistical fields that require updatings. The authors indicate that, in future, it may be possible to use social media data and apply more sophisticated analysis such as multimodal analysis to derive more about the dynamics of customer emotions.

This study by Bhutekar el al.[11] (2012) deals with the machine learning based approach to recognition of emotions in text, with particular contribution being made on such issues as feature extraction and classification. They used a children story corpus that are annotated with emotional labels and trained models using the supervised models like SVM and achieved greater levels of accuracy by reducing the feature sets, which included the use of features like Mutual Information, to determine the keywords that relate to a specific emotion. The study also contrived text and speech analysis module to offer a comprehensive method to emotion recognition. These results indicate that these methods are applicable on tasks such as text-to- speech applications where proper emotion mapping is essential since it is needed to produce subtle prosodies. In spite of this, the study admits the lack of analysis based on ambiguous emotional keywords, which were countered using strong validation. The work has useful suggestions on where to take affective computing and human-computer interaction further.

Bharti S et ai.[12] brings a two-fold algorithm to text-based emotion recognition that takes into consideration deep learning models (CNN and Bi-GRU) and machine learning technique (SVM). The

proposed model was applied to a mixed data set of sentences, tweets and dialogues with its accuracy being 80.11%. This works better than standalone machine learning and deep learning.

In a comprehensive review by Zidan M et al. [13], several emotion models, methods in which emotions can be detected and current developments, are explored in great detail. The growing importance of identifying feelings in text because of the rise of emotional data in social media is also pointed out in the paper. It talks about traditional machine learning as well as deep learning system of emotion recognition, and cites their advantages and limitations. The review also makes a comparison of the performance of the various models and has concluded that both traditional machine learning and deep learning-based methods are successful in solving the problems of textual emotion detection.

This article by the team of Karthikeya Gogineni et al.[14] (2023) hypersetils a new deep learning model designed to be efficient in sentiment analysis, including LSTM, GRU and CNN models, as well as various combinations of them as well (e.g. CNN+LSTM and LSTM+GRU). It uses such methods of feature extraction as Bag-of-Words and TF-IDF, but the latter is usually more accurate. The models are arranged in a text preprocessing, feature extraction, and a neural network-based classification pipeline. Four datasets, such as Movie, Coursera, Google play store and Flipkart reviews were considered consisting of binary sentiment labels. Of all the tested models, the CNN+LSTM performed the best on all the datasets with TF-IDF features. It can be seen in the results that the hybrid deep learning models achieve better results in comparison to the established machine learning algorithms especially when applied to a large dataset and the models exhibit high potential of being applied in large-scale automated sentiment texts.

Article by N Murthy G et ai. The article by [15] (2020) considers Long Short-Term Memory (LSTM) networks in the context of sentiment analysis of text data. The authors raise the issue of the importance of the sentiment analysis of the huge volumes of unstructured text that is produced every day. They pay attention to LSTM, a kind of Recurrent Neural Network (RNN) to use it in sentiment classification due to a possibility to cope with long-term dependencies in text. To conduct the test, the research uses the IMDB movie review and Amazon product datasets to perform the experiment of sentiment classification by employing LSTM.

Bruna O et al. The paper in [16] gives a review of different models of emotions used to classify texts into emotion set category, dimensional and the extended models. It implies that more personalized models, ones that consider personal traits, can assume more importance when the interaction between human and the computer is more personalized and where the computer has to be more aware of emotions.

This article is by Ziyi Yang et ai. [17] this paper deals with sentiment analysis of movie review data using deep learning technology. The researchers have used a corpus of film critiques characterized by a wide scale of movie reviews that incorporate textual content of critiques and ratings that would also act as sentiment marks of training and validating the models trained. The key model to be used is deep learning sentiment analysis model, in this case, a compound of Doc2vec to represent text vectors and Bidirectional Long Short- term Memory (Bi-LSTM) as the sentiment classifier . The data was evaluated with this integrated model besides other customary machine learning models which included Naive Bayes and Support Vector Machines, and another deep learning model, Convolutional Neural Networks (CNNs) . It has been revealed that the proposed Doc2vec+LSTM model produced high performance

with the result that the accuracy, precision, recall, and F1 values were 0.875, 0.870, 0.870, and 0.870 respectively . This shows that deep learning models, specifically the Doc2vec+LSTM model is more effective in sentiment analysis tasks involving movie movie review text than the conventional approaches because of their advantage of being able to take in the complex features representations and depict long-distance dependencies in text.

This research by Barik et ai. The article [18] suggests the use of a deep learning framework LSTM-DGWO to analyze the sentiment based on online customer reviews. The model combines the Bidirectional Encoder Representations through Transformer (BERT) model to embed words efficiently, the Genetic algorithm (GA) to extract features and the firefly algorithm (FA) to find the best features. The hyperparameters of the Long Short-Term Memory (LSTM) model is optimized using the Differential Grey Wolf Optimization (DGWO) algorithm, including learning rate and batch size. We assessed the model against a Shopify app store review data taken off Kaggle, having 50,140 merchandise -set reviews that were chosen randomly. The findings indicate that the developed LSTM-DGWO model produced a high accuracy score of 98.89 percent and a very low loss of 0.0484 that render the proposed algorithm better than traditional methods of sentiment analysis. .

The article by DR.Attili et ai. [19] adds the Nonlinear Deep Learning Framework Sentiment Classification Vector (NPSC) Algorithm, to improve sentiment analysis of textual information. The model has been built on a range of sophisticated structures such as BERT embeddings, to convert a word into a dense representation, Bidirectional LSTM (BiLSTM) to capture contexts in both directions, a Convolutional Neural Network (CNN) that extracts important local features, and attention mechanism that identifies meaningful words . Such modules are merged in a nonlinear fusion layer and the overall sentiment prediction would be performed through a softmax layer. This model was conducted to test the NPSC model on the IMDB movie review dataset, which has 50, 000 movie reviews, where positive sentiments are balanced with negative sentiments. The results of the experiments proved that NPSC performed much better in comparison with such modern models as RMDL or CLM in terms of different parameters and scores, reaching the accuracy of 94.2%, preciseness of 93.8%, recall of 94.5%, and the F1-score of 94.1% . Using ablation studies it was greatly demonstrated that each of these components played an important part in the performance of the NPSC by significantly dropping performance as one of these layers were cut out .

Li et al. [20] (2018) designed the BiGRULA model that combines Bidirectional Gated Recurrent Unit (BiGRU), lda2vec (a topic-enhanced word embedding model), and attention mechanism to provide better results in sentiment identification, both topical and sequential context. In the research, the evaluation was conducted with a variety of datasets such as IMDB, 20 NewsGroup and further real-life data of hotel reviews to assess domain adaptability of a model. In their study, it was revealed that their BiGRULA model performed better than the baseline models as its accuracy was 89.4% which was 3 percent ahead of the best existing approaches that was a sign of a good performance in tourism related sentiment analysis.

Chen et al. [21] (2023) proposed a dual-channel hybrid based on BERT with the integration of CNN and BiLSTM and an attention mechanism. The model utilises BERT embeddings in its input representation, CNN in finding local semantics and BiLSTM in global context. To test their BERT-based CNN-BiLSTM-Attention model, the study relied off of a dataset of hotel reviews, the size of which, however, was not explicitly stated, as well as its source. After their experiments, Chen et al. determined their

hybrid BERT-CNN-BiLSTM-Attention model to perform better with an accuracy of 92.35% and F1-score of 91.59%.

A recent study by Cahyaningtyas et al.,[22] (2021) suggested the Aspect-Based Sentiment Analysis (ABSA) model, which applied different deep learning methods, such as RNN, LSTM, GRU, BiLSTM, Attention BiLSTM, CNN, CNN- LSTM, and CNN-BiLSTM. To prove these claims, they compared these models on classification of the aspect and sentiment on Indian hotel reviews. The analysis-sample gathered 5,387 reviews of hotels in Indonesia using data taken in Agoda, booking.com, traveloka, etc. The set of data were identified and classified into the six aspects namely price, hotel, room, location, service, and restaurant. In this study, LSTM has been shown to be best at classifying the aspect with an accuracy of 92.6 percent, whereas CNN showed the best result in sentiment classification with the accuracy of 90.4 percent.

2.3 Summary

The reviewed papers explore a wide spectrum of traditional and deep learning approaches for sentiment and emotion analysis across diverse domains such as product reviews, tourism, customer feedback, and low-resource languages like Bengali. Several studies, including those by Suri et al. and Soundearajah et al., applied classical models like Logistic Regression and SVM, achieving moderate accuracies (e.g., 86.96% and 85.99% respectively), while Sallam et al. and Patel et al. highlighted the superiority of transformer-based models like BERT, achieving up to 93.67% accuracy in emotion classification. Hybrid and deep learning architectures, such as BiLSTM, CNN+LSTM, BiGRU, and their attention-augmented versions, consistently outperformed traditional models, as shown in works by Khan, Wang, Gogineni, and others. Notably, Li et al.'s BiGRULA and Chen et al.'s BERT-CNN-BiLSTM-Attention model demonstrated strong performance (up to 92.35% accuracy), combining topic modeling, attention, and sequential learning. Studies on low-resource datasets, like Hossain et al.’s Bengali review analysis using BiLSTM (91.35% accuracy), emphasize the adaptability of deep models even in constrained settings. Advanced optimization and fusion techniques, as seen in LSTM-DGWO and the NPSC model, pushed accuracies as high as 98.89% and 94.2% respectively. Collectively, these works affirm the effectiveness of deep and hybrid architectures, especially with attention mechanisms and transfer learning, in enhancing sentiment and emotion analysis across varied datasets and applications.

Table 2.1: Summary Of Related Work

Sr. No.	Authors	Dataset	Implemented Methods	Result
01.	S. Suri, K. Sharma, and S. Papneja (2020) [1]	Amazon fine food review	Logistic Regression	Accuracy:86.96%
			Naive Bayes classifier	Accuracy:85.7%
			KNN	Accuracy:66%
02.	AA Sallam, WQ Saif, TH Rassem, BA Mohammed (2024) [2]	ISEAR and AIT- 2018	BERT	Accuracy:93.67%
03.	Prince Patel, Dhara Patel, Dhruv Patel, Madhvi A. Bera (2023) [7]	labeled dataset	BERT	Accuracy:90%
04.	Eftekhar Hossain , Omar Sharif , Mohammed Moshiul Hoque, and Iqbal H. Sarker (2020) [8]	8,435 Bengali restaurant reviews	BiLSTM and word2vec embedding	Accuracy:91.35%,
05.	Sahithya Soundearajah; P. P. G. Dinesh Asanka (2022) [10]	ASOS product reviews	Nu-Support Vector classification model	Accuracy:85.99%
06.	Santosh Kumar Bharti, S Varadhaganapathy, Rajeev Kumar Gupta, Prashant Kumar Shukla, Mohamed Bouye, Simon Karanja	Mixed dataset comprising sentences, tweets, and dialogues	CNN and Bi-GRU	Accuracy:80.11%
07.	Mantasha Khan and Ankita Srivastava (2024) [3]	Twitter dataset	Logistic Regression, SVM,RNNs, and CNNs	-
08.	Tulika Chutia, Nomi Baruah (2024) [4]	-	LSTM , Bi- LSTM and BERT	-
09.	Xingyu Wang (2024) [5]	-	CNN,RNN and LSTM	-
10.	Pansy Nandwani, Rupali Verma (2021) [6]	SemEval, SST, and ISEAR	-	-
11.	Amjad Iqbal , Rashid Amin , Javed Iqbal , Roobaea Alroobaea , Ahmed Binmahfoudh and Mudassar Hussain (2022) [9]	Amazon reviews, IMDB, and Yelp	LSTM	-

12.	Swati D. Bhutekar, Manoj. B. Chandak (2012) [11]	Corpora	SVM	-
13.	Mahinda Mahmoud Samy Zidan Mahinda Mahmoud Samy Ahmed Zaki Zedan, Ibrahim Elhenawy ,Ahmed R. Abas,Mahmoud Othman (2022) [13]	-	-	-
14.	Karthikeya Gogineni AKiran Sai Reddy SKakarala HChowdary Gavini YPavana Venkat MHajarathaiah KKrishna Enduri M [14]	Movie, Coursera, Google Play Store, and Flipkart reviews	GRU+LSTM With bag-of-words GRU+LSTM With TF-IDF CNN+LSTM With bag-of-words CNN+LSTM With TF-IDF	52.25%, 94.29%, 60.49%, and 82.42% 52.0%, 95.34%, 62.17%, and 87.88% 82.50%, 95.35%, 74.16%, and 92.25% 81.36%, 95.55%, 74.80%, and 93.45%
15.	Dr. G. S. N. Murthy, Shanmukha Rao Allu, Bhargavi Andhavarapu,Mounika Bagadi,Mounika Belusonti (2020) [15]	IMDB movie review and Amazon product datasets	LSTM	-
16.	O. Bruna, H. Avetisyan, J. Holub (2016) [16]	-	-	-
17.	Ziyi Yang(2024) [17]	Movie reviews	Naive Bayes, Support vector machines, CNN and Doc2vec+LSTM	0.810%, 0.845%, 0.865 %and 0.875%
18.	Kousik Barik,Sanjay Misra , Ajoy Kumar Ray, and Anthony Bokolo .(2023) [18]	Shopify app store reviews	LSTM-DGWO	98.89%

19.	DR.Attili Venkata Ramana, Dr. Kalli Srinivasa Nageswara Prasad, Annaluri Sreenivasa Rao [19]	IMDB movie review	BERT, BiLSTM, CNN, and attention layers	
20.	Qin Li, Shaobo Li, Jie Hu ,Sen Zhang and Jianjun Hu (2018) [20]	ChnSentiCorp-Htl-unba hotel review, a set of Chinese hotel reviews	lda2vec ,bidirectional recurrent neural network model with attention mechanism	0.894%
21.	Na Chen ,Yanqiu Sun ,Yan Yan (2023) [21]	Chinese hotel reviews	BERT-based CNN-BiLSTM-Attention layer	92.35%
22.	Siwi Cahyaningtyas1, Dhomas Hatta Fudholi,Ahmad Fathan Hidayatullah (2021) [22]	Agoda,booking.com, traveloka	LSTM and CNN	0.926 % and 0.904%.

CHAPTER 3
METHODOLOGY

3.1 Introduction

This paper suggests a hybrid deep learning model capable to combine the lexicon-sharpened emotional features with context-aware neural modeling to effectively recognize frustration in online reviews. The method acknowledges the shortcomings of the conventional techniques of analyzing sentiment when analyzing more complex forms of feelings such as frustration by incorporating symbolic emotion signals alongside data-dependent situational insights. In particular, the description of affective features of the lexical level is based on emotion scores of the NRCLex lexicon and the sentiment polarity of the TextBlob. The features are then combined with deep contextual models by getting deep contextual representations via a Bidirectional Simple Recurrent Neural Network (BiRNN), which models both forward and backward dependencies in texts. In order to improve interpretability to better understand the most important phrases conveying emotions, a specific attention mechanism is provided, which makes the model focus on words that are most likely to reflect frustration. Moreover, the pre-trained word embeddings (GloVe) are used in order to enhance the semantics of the input text, whereas the data augmentation and class weighting techniques are utilized to mitigate the class imbalance. This unified approach attempts to establish a sound basis of identifying frustration, which will overcome the challenge of making the bridge between a symbolic concept of emotions and profound semantic learning.

3.2 Data Description

The original dataset comprises 1,048 rows and a single primary column labeled "reviewText", where each entry represents an individual online product review submitted by a user. Some reviews are collected from different people as primary data which number is 70.

Other reviews are collected from e-commerce platforms or similar online marketplaces like Amazon,Ali Express etc, where customers share their experiences regarding purchased products, delivery services, or overall shopping interactions. The textual content of the comments varies in length, tone, and style, ranging from concise one-line statements to multi-sentence narratives.

Although the dataset contains only the textual column (reviewText) as visible input data, each review is associated with a sentiment class label indicating whether the expressed opinion is positive, negative, or neutral. These sentiment labels provide a general emotional orientation of the feedback and are essential for identifying patterns that may indicate user frustration or others emotions like happy, sad etc.

	A	B	C
	Form_Responses		
1	Timestamp	Score	Write your valuable opinion about the online product that you recently purchased.
2	17/03/2025 20:58:09		Great quality for the price—definitely worth it!
3	17/03/2025 20:58:13		Delivered on time and performs as advertised.
4	17/03/2025 21:07:10		The speed is okay, but I was hoping for better performance.
5	17/03/2025 21:45:59		Delivered on time and performs as advertised.
6	17/03/2025 22:05:40		No problems encountered.
7	18/03/2025 00:37:37		I never purchase any online product
8	18/03/2025 01:16:55		No problems encountered.
9	18/03/2025 08:45:49		A great choice for those who need extra storage space.
10	18/03/2025 14:18:19		No problems encountered.
11	18/03/2025 19:46:08		No problems encountered.
12	18/03/2025 21:36:43		This product has been fantastic! I previously used a different brand, but this one is much better.
13	18/03/2025 22:19:44		Great quality for the price—definitely worth it!
14	18/03/2025 22:48:44		Delivered on time and performs as advertised.
15	19/03/2025 21:34:20		Bought this for my gadget, and it functions exactly as expected.
16	20/03/2025 10:20:02		A solid product that meets my needs perfectly.
17	20/03/2025 16:28:45		Delivered on time and performs as advertised.
18	22/03/2025 19:07:16		Didn't meet my expectations, so I had to send it back.
19	22/03/2025 19:09:47		No problems encountered.
20	22/03/2025 19:17:04		A solid product that meets my needs perfectly.
21	23/03/2025 01:47:51		Didn't meet my expectations, so I had to send it back.
22	23/03/2025 11:19:49		No problems encountered.
23	08/05/2025 22:40:35		No problems encountered.
24	08/05/2025 22:43:47		Delivered on time and performs as advertised.
25	08/05/2025 22:49:30		Great quality for the price—definitely worth it!

Fig 3.1 : Primary Data

	A
1	reviewText
2	Although the product itself appears to be working, the overall service was atrocious. The delivery was very slow, there was no communication and the item was poorly wrapped in a standard envelope. I would definitely NOT use this company again.
3	Poor quality, one connection didnt work. Several messages to seller, no reply and still waiting. Therefore not good value. Looked good thats about it. Would i buy from again ? NO !!
4	Item arrived used, not new, as was said in the description.
5	Second order with them, they got both orders wrong, they don't always answer the complaints to put their mistakes right, told them i would never order off them again, just terrible service off them. Dave M
6	Bad communication and advertising item for one price and as soon as you order, putting the price up even when you paid in full. I have meat up my credit rating because of the cancel the order when I all went through.
7	Paid for two only sent me one
8	Light not as advertised cheep Light was sent. Told to sent back at my expense, not worth the hassle. Do not buy this product
9	Cover wouldn't recognise fingerprint to unlock. Seller said they couldn't help. Very poor customer service.
10	Very very disappointed with the service and order ! The item arrived with the card missing and only adapter left . I contacted the seller and offered no replacement or refund. I actually got no response to last message. This is the first time in many years on ebay i had such an issue , so please be warned before placing an order .
11	The seller sent the wrong item. I ordered a battery pack phone case and they sent me a cheap phone case instead. Waiting for a refund. Not happy.
12	Not as described
13	Poor quality. The end of the charger broke off in my phone as soon as I plugged it in. I contacted the seller and they tried to blame me saying I used force and they have never had a complaint. Not even an offer to resolve the issue or replace it. Donâ€™t waste your money on poor quality products from this seller.
14	Very poor quality product. I sent a message to the seller telling him about the problem with the headphones. He told me to return it without going through eBay. I said I didn't have time to go to the post office because it's too busy for me. If I had time, I would go to the physical store.
15	I received only one of two creams I have ordered,still waiting for a response,poor communication,poor pack.aging the parcel,not happy
16	Not great, I got sent a child's toothbrush instead which I see advertised at a less price. I was happy for them to refund the difference but now they ignore. Don't buy of this seller!!!
17	Connection has broken and will no longer charge. I have contacted the seller via message but I have had no response after 4 days. Not impressed Otherwise quick dispatch time
18	Send me a used one full of scratches, no box,contact seller no response at all,make a favor and keep your money ðŸ˜‘
19	Product received one week after purchased, seller updated wrong tracking number to make me fool that Product dispatched. Be honest.
20	The product isnâ€™t good at all it snapped and broke my phone seller never responded!
21	Seller sends different caseâ€™s to whatâ€™s advertised!! Disgusting customer service avoid
22	Faulty product, seller just ignored my messages after I had returned it, luckily Ebay had my back. Avoid this seller.
23	I donâ€™t receive this item please send me this item
24	They sent a defective cable. It is impossible to charge the iPhone.
25	I will not be using this company again. I tried making contact so many times and they just ignored me. Stay clear.
26	Poor quality . Doesn't trim nose hair . Very cheap
27	Never received. Sent to the correct address first time. Royal Mail apparently lost it. Then delivered to the wrong address and never received even though requested a refund.
28	item never arrived seller stopped replying
29	Still not arrived
30	Sent me wrong item and was not like the one in the picture
31	Bad bad bad 0 response from seller
32	(SCAMMER) This seller is the worst kind of human avoid buying anything from them they will scam you for Â£4 if they can. He sells nothing but faulty goods. SCAMMER!!!
33	Item is in bad quality. Delivered late. I put in charge , when i switch on, is not working at all
34	Item was faulty, messaged seller and received no response.
35	Item never arrived messaged a few times got ignored, so had to ask eBay to step in to get me a refund
36	Doesnâ€™t actually work! Cheap rubbish
37	Broke after two days ,absolutely rubbish, please people don't buy it , it's cheap rubbish and changes nothing properly.
38	Product was faulty. The seller said he would provide a return label but this didnâ€™t happen. I wouldnâ€™t recommend this product.
39	I couldn't get this mouse to work. i tried it with three different laptops with no luck.
40	The seller did not bother to provide instructions for use. In the end, no matter how I tried to turn on the device, no matter what combinations I used, everything turned out to be useless. It was probably damaged

Fig 3.2 : Secondary Data

3.3 Applied Model Overview

The main purpose of this research paper is to detect frustration on customer product reviews using deep learning.Different preprocessing techniques are used here and train the model.Then the model performance is estimated here .All the steps are discussed briefly in this chapter and also shown in a workflow diagram.

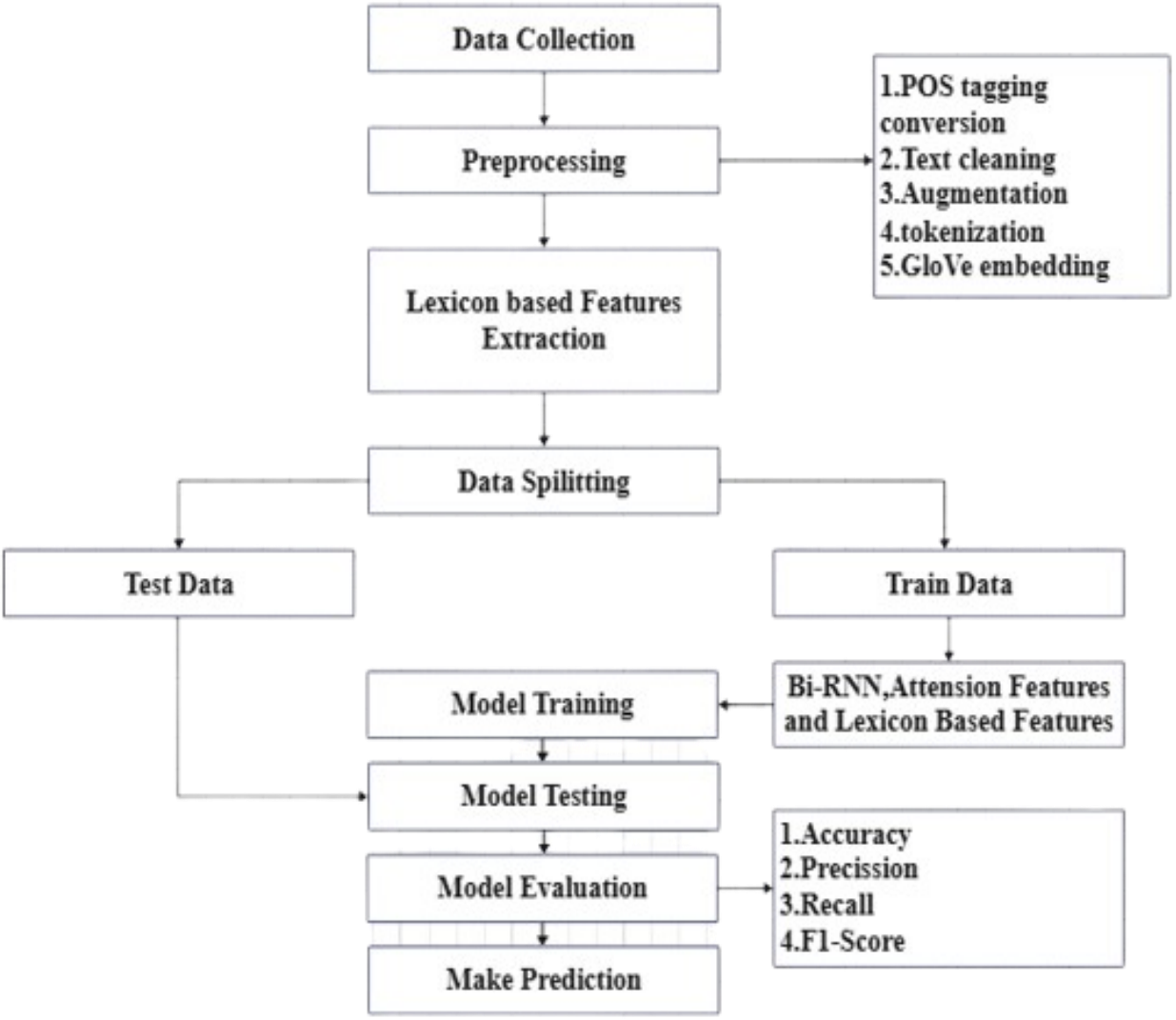

Fig 3.3 : Work Flow Diagram

3.3.1 Data Collection

The first phase of any natural language processing (NLP) task is called data collection in which raw textual data is obtained that represents different data sources within the scope of the problem under investigation. This data, in the case of frustration detection/ sentiment analysis, may be gathered by

means of online reviews, customer feedback and reviews, social media posts, forums, or other products review sites, such as Amazon, eBay,Ali express or TripAdvisor. It is aimed to get an opportunity to get a representative and diversified sample of data reflecting various manifestations of user experiences and emotions. The partner data collection is effective in exposing the model to a diverse use of language, word usage, and emotional response information that the model needs to develop a good and generalizable NLP model.

3.3.2 Data Preprocessing

One of the steps that take raw, unstructured text and prepare it in a more structured manner that could be inputted in a machine learning or a deep learning model is data preprocessing. It encompasses some of the most important procedures which are removal of noise (such as special characters and numbers), lowering language, removal of stopwords (such as is, and, the), and lemmatization that reduces words to their basic forms. There is also tokenization that is done to divide sentences into each word or tokens. Further preprocessing can also include POS tagging, dealing with spelling errors, extracting sentiment or emotion features with a tool such as TextBlob or NRCLex. This step is important since quality and unified data will enhance and raise the comprehensiveness and productivity of learning. Part-of-Speech (POS) tagging is the replacement of every word in a sentence with its grammatical role, e.g. noun, verb, adjective, or adverb. In particular, when lemmatizing the words converting POS to tagging may be of practical value i.e., making a word to its de-root stage. Various parts of speech contain different lemmas thus proper POS tagging means a better normalization of the text concerned. To give an example, the fly word could be lemmatized as fly when it is a verb or remained as flies when it is a noun. During this process, POS tags retrieved using tools such NLTK get matched with similar tags in WordNet and they help lemmatizer in determining the correct form. This contributes massively to the downstream tasks of detecting sentiments or emotions to a great degree as they can be assured that the words will be represented properly.

For example, consider the review from the dataset:
"Poor quality, one connection didnt work. Several scratches on the product."

After preprocessing:

1. **Lowercasing & Noise Removal:**
   "poor quality one connection didnt work several scratches on the product"

2. **Tokenization:**
   ["poor", "quality", "one", "connection", "didnt", "work", "several", "scratches", "on", "the", "product"]

3. **Stopword Removal:**
   ["poor", "quality", "connection", "didnt", "work", "several", "scratches", "product"]

4. **POS Tagging:**
   [("poor", JJ), ("quality", NN), ("connection", NN), ("didnt", VBD), ("work", VB), ("several", JJ), ("scratches", NNS), ("product", NN)]

5. **Lemmatization With POS:**
   ["poor", "quality", "connection", "do", "work", "several", "scratch", "product"]

3.3.3 Data Cleaning

An essential data analysis or machine learning pre-processing step is data cleaning. It entails the process of detecting and eliminating oversights or discrepancy within the data in order to maintain quality and credibility of data. Some of the steps involved in the process are: eradication of duplicate records, management of missing values, fixing inconsistent formats, deleting redundant or noisy information and standardization of text (e.g. converting everything to lowercase, removing the punctuation marks, or the expansion of contractions). Among text data, text data cleaning involves the deletion of stopwords, special characters, HTML tags, Stemming or lemmatizing to bring words to their root word form. Good data cleaning makes the resulting downstream task more accurate and robust because it minimizes the noise, and because the input data is consistent, significant, and open to modeling. Most advanced algorithms will yield inaccurate or less than optimal data with outgoing in depth data cleaning. For example, from the dataset:

Before Cleaning

"Poor quality. The end of the charger broke off in my phone as soon as I plugged it in. I contacted the seller and they tried to blame me saying I used force and they have never had a complaint. Not even an offer to resolve the issue or replace it. Donâ€™t waste your money on poor quality products from this seller."

After Cleaning:

"poor quality end charger break phone soon plug contact seller try blame say use force never complaint even offer resolve issue replace dont waste money poor quality product seller"

3.3.4 Data Augmentation

In natural language processing (NLP) the process of data augmentation is used to add synthetically generated text samples to the training data to make it bigger and more diverse. It is especially valuable in the case of class imbalance - some labels (e.g., "frustrated") can be seen significantly less often than others. Synonym replacement is one of them which involves choosing some words in a sentence and substituting with their synonyms with the use of a lexical database such as WordNet. As an example, instead of saying- The service was slow, you can use- The service was sluggish. Augmentation will avoid overfitting the model by creating similar meanings of new samples and enabling the learner to generalize unseen data more specifically when the nature of emotional categories is underrepresented in the training data.

For example, from my dataset:

Before Augmentation:

"Very very disappointed with the service and order! The item arrived with the card missing and only

adapter left. I contacted the seller and offered no replacement or refund."

After Synonym Replacement:

"Extremely dissatisfied with the service and purchase! The product arrived with the card absent and only adapter remaining. I reached out to the vendor and received no substitute or reimbursement."

3.3.5 Tokenization

A tokenization corresponds to the transformation of a sentence into single components called tokens or tokens - usually words or subwords. In deep learning pipelines, subdivision of words into tokens is followed by the transformation of these tokens into numerical sequences that the model reads. This is achieved through the use of a tokenizer which replaces a high frequency/appearance word with an integer. The resulting list of integers is then padded to attain fixed length to ensure even input size. Tokenization helps to connect to the process of modeling raw texts into numbers so that it can be processed in neural networks.

An example of the sentence like: An absolute fake, would be broken down to [“An”, “ absolute”, “fake”].

3.3.6 GloVe Embedding

Global Vectors for Word Representation is a pre-trained word embedding algorithm and represents each word in the vocabulary as a fixed-length dense vector usually of dimension 100,200 or 300. These vectors represent semantic links among words on the ground of their commonness with other words in sizable text corpora. As an example, the term king and queen will share vectors that will be near one another in the space. In the deep learning models, GloVe embeddings are used to initialize the embedding layer enabling the model to begin with a wealth of knowledge about the meaning of the words. This enhances model performance because of the limited need to learn word semantics de novo and when the amount of labeled data is few.

```
Total words in GloVe embeddings: 286585
Shape of embedding matrix: (10000, 300)
```

	0	1	2	3	4	5	6	7	8	9	...
0	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00000	0.00000	...
1	0.000000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00000	0.00000	...
2	0.104910	-0.023204	0.35099	0.107130	0.370400	0.263460	0.023988	0.075262	-0.45840	-0.61076	...
3	-0.836760	0.022640	-0.26628	0.066996	-0.080808	-0.059011	-0.403300	-0.647690	0.13826	-1.85360	...
4	0.071531	0.298400	-0.24760	-0.020701	-0.247870	-0.125450	-0.461070	-0.362510	-0.14852	-2.00940	...

5 rows × 300 columns

Fig 3.4 : GloVe Embedding

3.4 Lexicon Based Feature Extraction

In natural language processing Lexicon based feature extraction is a technique in which text- derived features are made using pre-determined dictionaries, also called lexicons. These assign a string of words a score to indicate sentiment or emotion. Every word in the text is compared with those existing in the lexicon to find out the sentiment associated with it (good, bad, or neutral), or category of emotion (joy, anger, sadness, etc.). Features such as total sentiment score, the number of positive or negative words or frequencies of emotion can be derived based on these matches such that tasks like sentiment analysis or emotion detection can use these features. The approach is interpretable, coded and does not require labeled training data but can be less effective with context, sarcasm, and domain-specific vocabulary.

3.5 Dataset Splitting

The process of dataset splitting is one where the dataset will be subdivided into two sets: training and testing (sets). Training set is applied to construct and train the model whereas testing set is applied to estimate the performance of the model. This process can be assured that the predictions of the model and its generalization ability can be tested.

Training Dataset: This is the bigger part of the dataset that will be applied in training the machine learning. It assists the model to acquire patterns, relationships, and features using the data to come up with accurate predictions.

Testing Dataset: The tests data are kept aside the training data. It is applied to evaluate the level of performance of the model on new data. These test set results give an approximation of the success that the model will have in practice.

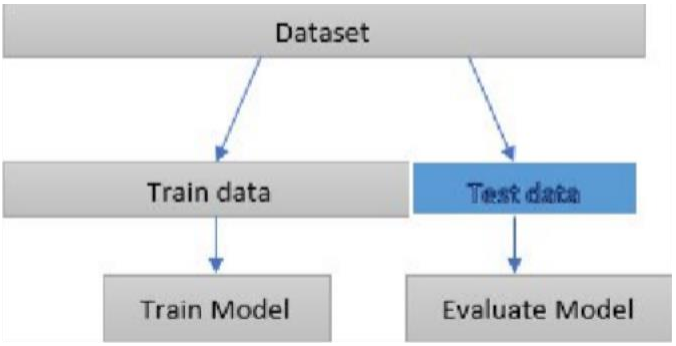

Fig 3.5 : Data Splitting

3.6 Bi-RNN, Attention Layer And Embedding

Attention Layer

A attention layer is a unit of a neural network which is able to dynamically attend on the most important features of an input sequence in training and inference especially on natural language processing tasks. As opposed to the standard layers where all time steps are treated the same way, the attention

mechanism enables the model to automatically award a varying level of importance scores to different tokens or words depending upon their contextual importance. This mechanism is coded manually in a custom attention layer, which uses the extend method, tf.keras.layers.Layer, and allows controlling and changing internal calculations. On this layer a weight matrix W and a bias b is initialized which learns to compute alignment scores that indicate the relevance each time step to the overall meaning of the sequence. The scores receive a non-linearity via tanh activation function and finally use a softmax function to normalize the scores into a probability distribution. These arising attention weights establish the stating of what has to be stressed or unstressed within the sequence. The attention weights are then multiplied with each input vector (e.g., the word representation of an RNN output), thus yielding a weighted sum of input vectors (i.e., the context vector). This last context-vector is a narrow representation of the sequence that contains its most important information to allow the model to make better predictions. The ability of the developers to customise the attention layer allows them to make any specific use out of it, such as being able to detect instances of frustration within a given review, whereby some words or phrases in a review can have far more emotive meaning than others.

Input Layers:

In the case of the deep learning process, input layer will be used as the base points at which data will be fed to the learning model so as to allow the various forms of data to be processed and learnt concurrently. This architecture specifies two input layers to process heterogeneous data sources differently: the first one is a sequence input layer, called text_input, that receives textual data tokenized and padded (that is, expressed as fixed length integer sequences) into its input. both sequences correspond to a user review or text sample with each word in the sample representing one index with respect to a vocabulary. This input will play an important role in obtaining the semantic and syntax structure of the text. The second input layer frust_input is to be fed with a vector of other hand-built features based on the lexicon-based emotion parsing such as frustration, anger or sadness scores, most of which are obtained via such tools as NRCLex or TextBlob. These characteristics add additional context in emotional sense to complement the model in recognizing the subtle emotions. The model is a multi-input neural network since both kinds of input, textual sequence data as well as emotion based features, are involved. It can learn both scattered linguistic content and organized emotional clues. This mixed input structure enhances the symbolic ability of the model, which eventually makes it perform better in the tasks such as detection of frustration that needs both context and emotion.

Embedding Layer:

An embedding layer is a primary feature of natural language processing models that maps discrete (i.e. one word corresponds to only one value index) word indices into continuous-valued vectors which are dense (i.e. have many values) and enable the model to find semantic relations between words. The input text that has been tokenized and arranged with each word taking on a unique integer ID, is embedded in space by feeding those integer IDs to the predefined embedding matrix to assign each of them a fixed size dense vector. Pretrained word embeddings, like Word2Vec or GloVe, trained on large text corpus and are filled with the semantic structure of the language are frequently used to initialize this matrix. All the vectors in the matrix are words in a multi- dimensional space with semantically related words located near one another. The trained layer can fine-tune these word vectors to be more task specific, e.g. frustration detection, as it is made trainable. This has the effect of being an interface between uninterpreted, symbolic language (words or tokens) and the numerical types that neural networks expect,

but allow the preservation of the meaning and context of the words, and can be efficiently processed by the rest of the network (e.g. RNNs or attention mechanisms).

Bidirectional Simple RNN layer

Bidirectional SimpleRNN layer is a variant of the recurrent neural network (RNN) based architecture that handles sequential data both forward direction as well as backward direction thus making the model have access to deeper contextual information. A typical RNN is able to go through an input sequence forward right-left, left-right, or right-left, only but that does not mean that it cannot learn patterns using the future. In practice, however, in most cases in natural language processing, where we are trying to detect the sentiment of a post or the level of frustration of a speaker, the meaning of the word or the combination of words may not be defined by previous words, but also the next words. This limitation is overcome by the bidirectional structure which executes two parallel SimpleRNN layers in which the first computes the sequence in the original (forward) direction and the second one in reverse (backward) direction. These are in turn concatenated and give a composite representation that indexes both directions of context. This two-eyed perception assists the model to take note of more intricate relationships in text. The parameter of return_sequences=True is defined in this architecture in order that the layer enables the output of a complete sequence of hidden states during each time step which is necessary to use the next attention mechanism. Through the use of both past and future context, Bidirectional SimpleRNN layer contributes to dramatic improvement of foreseeing the potential of the model in comprehending and categorizing subtle emotional patterns in textual data.

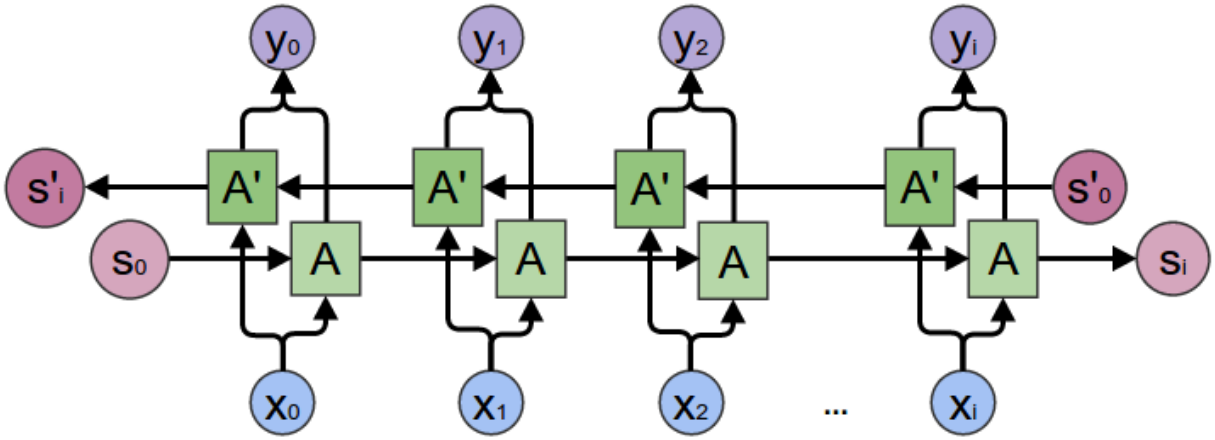

Fig 3.6 : Bidirectional RNN Model

Layer Normalization:

Layer Normalization is a normalization technique that has to be applied to the activations of a layer in a neural network to stabilize and expedite the training process. Layer normalization unlike batch normalization, which normalizes against the batch dimension and can be sensitive to the batch size; normalizes over the features dimension of each individual training example. That is, to each data point,

mean and variance get calculated on all the outputs of that layer and then the activations are normalised. In so doing, layer normalization avoids more layers: internal covariate shift, the incidence by which the distribution of inputs to a layer varies over time through training decelerates learning. This stabilization also results in more rapid convergence, and can in many cases enhance model generalization. Layer normalization works especially well in recurrent networks such as RNNs or its variants, where it is useful to process variable-length sequences, and not sensitive to variable batch sizes, which makes it suitable to sequential data. In the calculated model, Bidirectional SimpleRNN layer is followed by layer normalization to normalize hidden states that are later sent to the attention mechanism to make the learning process smoother and more consistent.

Dropout Layer:

Dropout is an effective regularization method employed in neural networks to guard against overfitting by randomly selecting or passing over the network numerous neurons by randomly dropping them or shutting them down after every training step or iteration. In particular, dropout temporarily removes a fixed percentage of nodes (along with their connections) during training, lest the model become over-dependent upon any particular neuron. This has the effect of encouraging the network to learn stronger, more distributed representations, since the rest of the active neurons have to make up and collaborate to make good predictions. Notably, dropout is used only during training; at inference time no dropout is implemented, rather the outputs of all neurons are scaled to match the training-time dropout rate. Dropout also facilitates the generalization of the model to unseen examples and more generally promotes model performance and decreases the likelihood of memorizing the training data by decreasing co- adaptation among neurons. In this model, the dropout blocks were inserted in the best positions after the output of the attention block and density rates of 0.4 and 0.3, respectively, were used to regularize the network in several places and make it more resistant and resist overfitting on the task of frustration detection.

Concatenation:

In neural networks, the concatenation operation of numerous tensors (vectors, matrices or higher dimensional arrays) along a given dimension fuses together their content to create one unified representation. This method comes in handy when a model requires mixture of the feature type or sources of data, such as aggregating of the deep learned features on a given neural layer with handcrafted or external features, such as lexicon-based emotion scores. Concatenating these feature vectors makes the model more informative and comprehensive since they give a variety of inputs that addresses a variety of data aspects. Practically, concatenation is used to layer the inputs on top of each other thus adding the dimensionality on an axis of choice without any mixing, or distorting of the entering elements per se. This enables the higher layers to discover more complicated interactions among the set features whereas enhancing the capacity of the model to represent delicate patterns. In your frustration detection model, concatenation is implemented between context vector generated by attention mechanism and the frustration lexicon features as auxiliary, so that the model can combine both deep semantic knowledge and explicit affective signals to conduct the purpose of better classification.

Dense Layer:

One of the simplest possible building blocks in neural networks is a Dense layer, otherwise called a fully connected layer. It inter-relates each of the neurons in the input with each of the neurons in the output, and the model learns complicated features in schema by integrating features in malleable and non-linear fashion. The Dense layer consists of connections that have matching weight which during the training process is updated to reduce the prediction error. The elements of the Dense layer are computed as a weighted sum of the inputs with a bias term to raise the non-linearity of the output and use the activation functions such as ReLU (Rectified Linear Unit) or sigmoid to find more complex mapping. The latter is especially strong due to its ability to model interactions among all input features and therefore it is very expressive. As in deep learning networks, Dense layers are normally at the output end of the model where they can represent decision-makers, converting the learned feature representations into end predictions - e.g., probabilities in a classification problem. Within your model, the Dense layer is a key component that is to really amalgamate the contextual information (which comes in as the result of applying the attention layer) with the lexicon-based features in order to produce a useful decision that a certain text implies frustrations or not.

Output Layer:

The last layer in a neural network is called the output layer which does the actual prediction of a model and takes in the information remained on all prior layers. Its organization and activation operation depends upon the form of the task, like regressions, multi-class classification, or finish classification. During processing with a binary classification where a component detects frustration, the output layer will usually include one neuron whose activation function is a sigmoid that will convert the final weighted signal into a response between 0 and 1. This value exhibits the likelihood the given input text falls under the class positive (e.g., frustrated). When output is near to 1, the model suggests that there will be frustration whereas when output is near to 0, there will be non-frustration. The output layer is a direct result of the extent which the model learned to differentiate between classes out of training data. It accepts feature representations that are at higher-levels, run through embeddings, RNNs, attention and dense layers and reduce the sophisticated understanding into one final decision. Output layer is simple in nature, unlike the layers preceding it, although its structure is vital, as it determines the interpretation of the network outcomes and their further application in evaluation or deployment of the results.

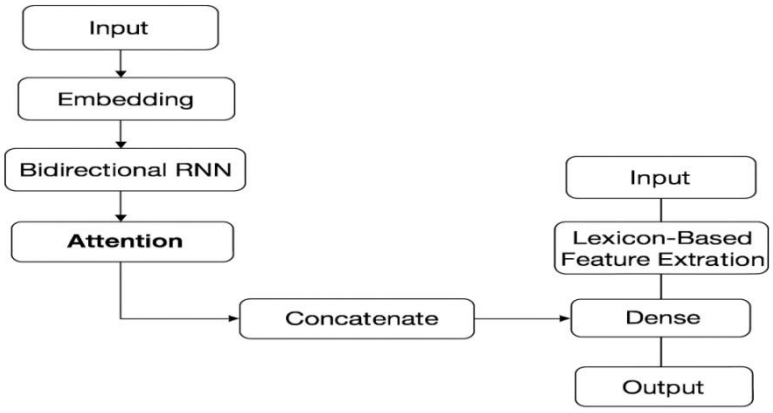

Fig 3.7 : Bi-RNN,Attention Layer and Lexicon Based Feature

3.7 Model Training

Training of a model is the process during which a machine learning or deep learning model is trained using data to find out the patterns and correlations between input and output. In training, the model takes batches of the labeled training data (feature) and creates predictions. There is a loss that is used to compare these predictions to the true labels and the loss is used in counting the error in the predictions. The model will then run an optimization process (e.g. Stochastic Gradient Descent or Adam) that updates its internal weights and biases via backpropagation, in an attempt to minimize loss. This is again repeated across many number of times called epochs where each epoch contributing to the incremental improvement in the model performance. Training may involve such methods as dropout, regularization, and early stopping, to prevent overfitting and enhance generalization. The result of model training is a set of parameters learned that best map on the training data that can be used to make accurate predictions on other, untrained data.

3.8 Model Testing

Model testing entails the assessment stage, which establishes the level of effectiveness exhibited by a trained model with respect to its performance on unseen data; otherwise referred to as test set. Such information is not trained and acts as a surrogate of real-world situations. The primary aim of testing is to determine the generalization capability of the model or in other words to test how accurately it will predict unseen data. The trained model will output results in the test set input and the output will be compared to a real label. The performance is then measured in terms of different evaluation measurers including accuracy, precision, recall, F1-score or AUC, etc. based on that. Good performance on test set is good evidence that a model has not only memorized the training data well but has also remained flexible enough to deal with variance that might occur on new data. It is a very essential step before an application of a model since it provides a model reliability and robustness in practice.

3.9 Model Evaluation

The step of model evaluation is a vital component of any machine learning or deep learning pipeline, especially the ones that imply sentimental analysis or classification of feedback. It establishes the extent to which the trained model capable of generalization of the unseen data and the ability to accurately classify practical examples. Models must be reliable, accurate and sensitive to linguistic nuances when models are used to classify customer feedback, and can determine whether they are frustrated, satisfied or sentiments. Here the evaluation metrics enter the game and provide us with numbing measures of how well a model works in various dimensions.

Accuracy:

One of the most frequently applied measures, in classification problems, is accuracy. It determines the capability of a model to correctly identify the incidences (both positive and negative) of all the instances predicted regardless of the model type. When a model can successfully actually predict 90 out of the 100 total samples, then that model would be deemed to have an accuracy of 90 percent. Although it gives a rough idea of how well the performance is done, this accuracy is misleading when dealing with unbalanced datasets. As an example, when a model is faced with a dataset in which 95 percent of the

entries belong to one class and the minority class, a model may predict anything as the majority class and still have a 95 percent accuracy even though it has failed at all in detecting the minority.

Precision:

Precision dwells on the quality of the positive predictions that may be related to the model. It has been defined to be the proportion of true positive predictions (i.e. correctly predicted positive cases) of total predictions made by the model as positive. In high precision, once a positive label is given by the model, then chances are high that the label is indeed correct. When the cost of false positives (or risk of catastrophic outcomes, as in the case of fraud) is high, precision is particularly important, e.g. in spam detection or fraud detection a low rate of false positives (or risk of the reputational damage caused by wrongly on the red list) is essential. Precision is not the same as accuracy as it factors in the positive predictions rather than the pool of the total predictions.

Recall:

Recall referred to as sensitivity or true positive rate is the number of all the positive cases identified by the model in the set of relevant predictions. It represents the divide of correct positives by the total correct positives and false negatives. High recall implies that the model is able to pick most or all of the actual positive items although some false negatives are sometimes classified as positives. The importance of recall applies to those instances when non-detection of a positive example is expensive like in detection of diseases or frustrated customers. In contrast to precision that measures the accuracy of the positive predictions, recall measures the coverage of the positive predictions.

F1-score:

F1-score is the harmonic mean of precision and recall and it is a single value that trades off both. It is especially applicable in case of trade-off between precision and recall, and one wants a measure that considers both issues. F1-score is also relevant in the situation where false negatives and false positives would equally be bad. It is between 0 and 1 with higher results showing the better performance. F1-score is unlike the other 3 due to the fact that it does not only concentrate on correctness (as is done with accuracy), or on one of the two constituents of positive prediction (accurate, precise or recall). Instead, it amalgamates the two measures of precision and recall into a single measure.

These metrics are thoroughly discussed to evaluate the models effectiveness.

Confusion Matrix

A confusion matrix is an illustration constituting a table where the performance of a classification model is measured in the form of correct and incorrect predictions of the model against the actual results. It has been split up into four groups such as true positives, true negatives, false positives, and false negatives. This matrix assists to comprehend the extent to which the model separates the classes and offers more ideas than a mere accuracy. It comes in specifically useful on ascertaining the errors of which type the model is erring, as well as estimating key parameters of performance like precision, recall, as well as F1-score.

3.10 Frustration Detection

Once trained, the model will classify whether a given review expresses frustration, not frustration. The classification could be binary (frustration vs. not).

3.11 Used Hardware And Software

This project requires several tools and libraries:

- **Python:** This is a powerful, general purpose programming language that allows the use of basic and advanced features and it suits beginners and advanced programmers.

- **Pandas:** A free Python data analysis and manipulation library written by Wes McKinney in 2008 that is based on a data frame and offers high performance. It is both amateur and professional friendly.

- **NumPy:** Python module meant to perform numerical calculations and the increasingly effective manipulation of one and multi-dimensional arrays. It is written mostly in C which allows computing apparatus to be rapid.

- **Jupyter Notebook:** A free and open-source interactive web-based notebook that allows construction and sharing of live code, mathematical expressions, visualizations and explanatory text. It is compatible with various languages such as Python, PHP, R, C programming language.

- **Matplotlib:** Package of multi-platform data visualization based on NumPy arrays. It was created by John D. Hunter in 2002 and facilitates plotting within Python scripts, web application.

- **Seaborn:** Seaborn is a library of statistical data visualization written in Python. It improves the look of the graphs with color schemes, and APIs depending on the publications of data and based on the Matplotlib.

- **ELI5:** A library of Python intending to explain and debug inferences of machine learning models. It applies such algorithms as Text Explainer and Permutation Importance to analyze black-box models.

3.12 Conclusion

Improving the accuracy of customer feedback analysis on different sites depend on the quality of data collection and the features used for model training. Early identification and prevention of issues in customer comment analysis can reduce administrative overhead. This chapter outlines the anticipated results and potential implications of the applied research, setting the stage for the discussion in the next chapter.

CHAPTER 4
RESULT ANALYSIS

4.1 Overview Of Results

Our suggested hybrid deep learning model, the combination of bidirectional RNNs with lexicon- based features and attention mechanism also showed great results in frustration detection, based on customer reviews. Over the training process, the model had been able to steadily enhance its accuracy of learning, with the chances of accuracy in the validation process finally achieving approximately 94.6%, with the test accuracy being impressive at 95.0%. The outcomes revealed that the unit adequately replicated contextual sentiment characteristics and emotional signs in textual data. The addition of emotion scores of the lexicon of NRCLex and features of polarity/subjectivity of the TextBlob improved the model in treating frustrated reviews, as opposed to non-frustrated, a great deal. Furthermore, the data augmentation, which balanced the minority classes, also allowed achieving good generalization. In general, the model showed better performance than other conventional classifiers and had a high level of reliability implying that it can be possibly used in practice in systems of monitoring a sentiment.

4.2 Accuracy Per Epoch

Throughout the training process, both training accuracy and validation accuracy were monitored to ensure that the model generalizes well beyond the training data. A small gap between these metrics typically suggests that the model has not overfitted and has learned features that apply to unseen data. Given the final validation accuracy of 95%, it is likely that the training accuracy remained closely aligned, indicating good generalization performance. Furthermore, the model's ability to achieve strong results on the test set reinforces that it avoided overfitting and learned meaningful patterns from the dataset. The application of regularization techniques such as dropout layers, early stopping, and learning rate reduction likely contributed to this generalization by preventing the model from memorizing the training data.

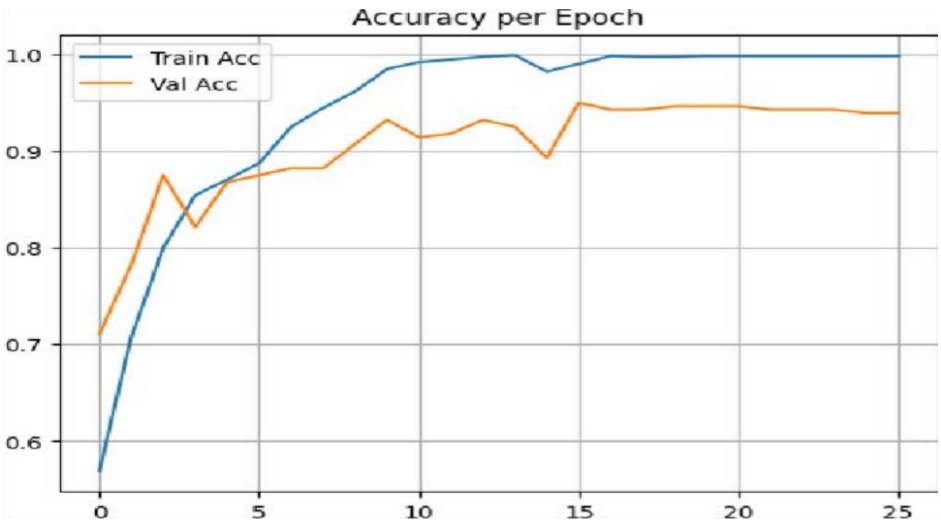

Fig 4.1: Accuracy Per Epoch

4.3 Performance Metrics

The classification report shows good results of the frustration detecting model. It indicates that this model has a precision of 0.94 and a recall of 0.96 to determine that the majority of the samples not frustrated is successfully found, and the false positive is a minimal number as well. In the case of Frustrated class, the precision is elevated to 0.96 with recall of 0.94 indicating that the model is also very good in spotting frustrated review not to mention missing some true frustrated reviews. Both have an equal F1-score of 0.95 which is a good indication of balancing precision and recall. On the whole, the model has an accuracy of 0.95 which is quite good, as well as the macro and weighted averages are the same, 0.95, so the dangers of bias towards a particular class are excluded.

Table 4.1: Classification Report

Class	Precision	Recall	F1-Score	Support
Not Frustrated	0.94	0.96	0.95	141
Frustrated	0.96	0.94	0.95	139
Accuracy			0.95	280
Macro Avg	0.95	0.95	0.95	280
Weighted Avg	0.95	0.95	0.95	280

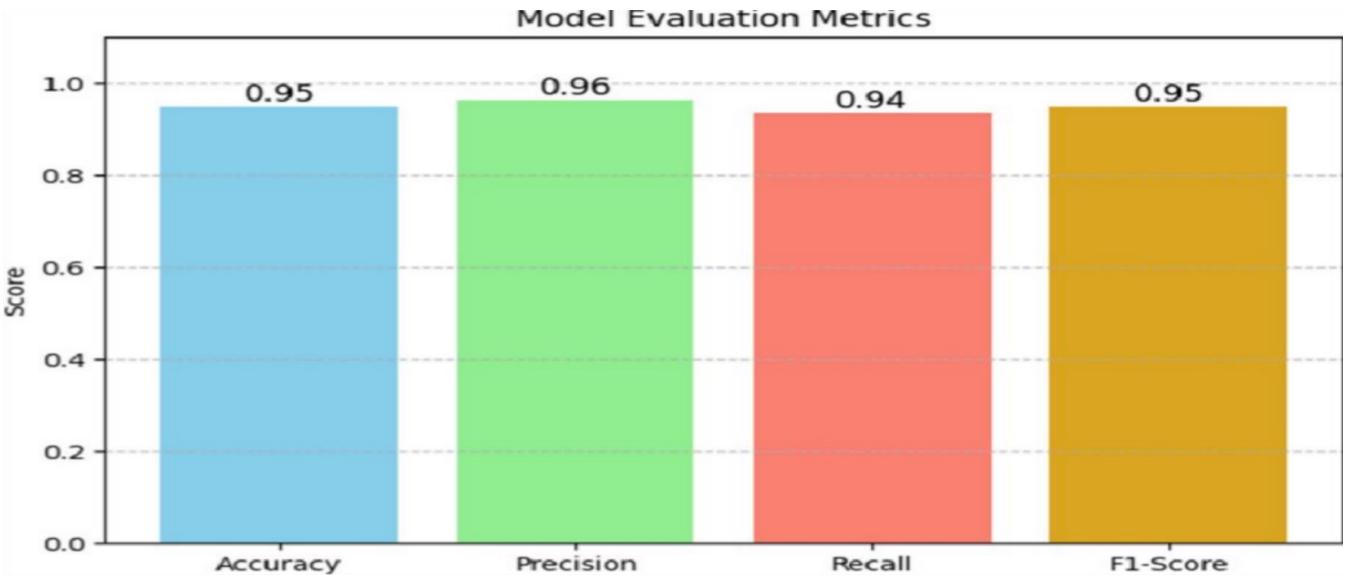

Fig 4.2 : Model Evaluation Metrics

4.4 Confusion Matrix

The confusion matrix (Fig 4.3) allows looking in details how the model performs in terms of classification by showing in how many cases it was correct or incorrect in terms of predicting a particular class. The model got 136 not frustrated reviews (true negatives) and 130 frustrated (true positive) reviews correct in this matrix. It had 5 false positive and 9 false negative predictions

respectively which means that non-frustrated reviews were falsely reported to be frustrated and frustrated reviews were falsely reported to be non-frustrated. Such distribution exhibits good sensitivity and specificity of the model that performs well and equally in both classes. The comparatively small amount of misclassifications further proves the ability of the model to reflect frustration-related patterns being present in textual data accurately.

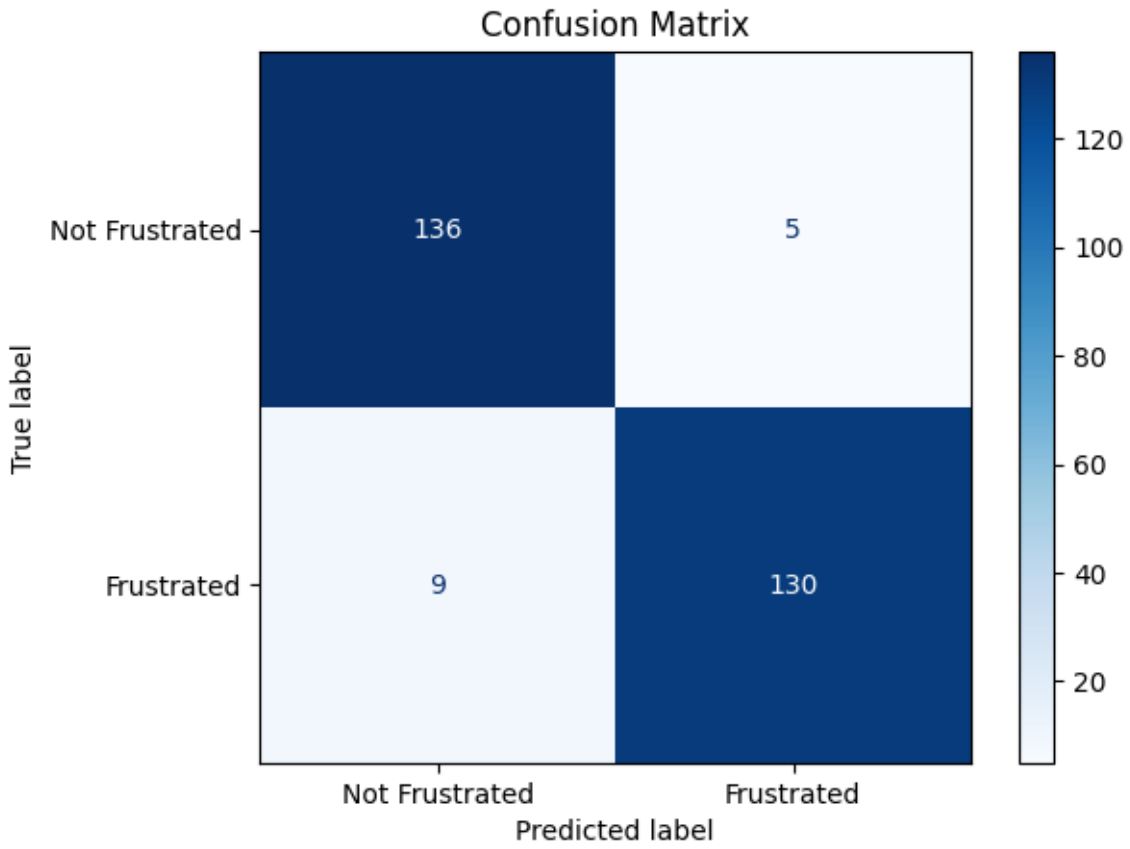

Fig 4.3 : Confusion Matrix

CHAPTER 5
CONCLUSION

5.1 Conclusion

This research provided a hybrid model that was a combination of Bi-directional Recurrent Neural Network (Bi-RNN) framework and attention mechanism and lexicon-based features implemented by such tools as NRCLex and TextBlob. It has been observed that the model was high-performing in terms of detecting frustration in customer reviews through its use of the contextual awareness property of Bi-RNN, the attention layer where the focus gatherings, and the emotional granularity of the features described by lexicons. With the attention mechanism, the model was able to focus on the sentiment-rich denominations of the text and the addition of extra features like polarity, subjectivity, and diversity in emotions played a prime role in enhancing the level of classification. Taken as a whole, the model was useful in distinguishing between frustration and other emotional states such as satisfaction, anger or neutrality, which means it can be an excellent tool in the arsenal of businesses striving to track and react to customer moods in real-time.

5.2 Limitations

Regardless of its good performance, the model has some shortcomings. First, it might have a problem with identifying frustration when reviews show mixed feelings or sarcasm, which are more than merely a simple understanding of language. Second, lexicon-based features may add value to the sentiment content but may also increase transaction noise when emotions are weakly or ambiguously done. Third, training was based on a relatively small set of data exclusively containing review texts; therefore, the model might not generalize well to the other communication medias like chat transcripts or social media posts. Also, since training deep models (such as Bi-RNN with attention) is computationally demanding, it might need critical setting of the hyperparameters to achieve the best results.

5.3 Future Scopes

There are some ways that future work can build on the model to make it more effective and generalize in the future. The tf/idf ranking could be improved by incorporating transformer- based architectures, e.g., BERT or RoBERTa, to have a better representation of more complicated linguistic patterns in a sentence, e.g., sarcasm or implicit emotion. By broadening the dataset to encompass multimodal data (e.g. audio, chat logs or video transcripts) the model may be both more cross-platform robust. Moreover, the expansion of the multilingual support would allow the model to operate on customer reviews in other languages. Probably, improving the performance of longer texts could be achieved by fine-tuning the attention mechanism or incorporating hierarchical attention. Lastly, incorporation of the system into pipeline of real-time feedback analysis would enable businesses to respond to customers frustration in real-time, therefore, making services responsive and end-user friendly accelerating service response time.

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CODE

```
# STEP 0: Install & imports
!pip install nrclex nlpaug textblob
import pandas as pd
import numpy as np
import re
import nltk
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras.layers import Input, Embedding, Bidirectional, LSTM, SimpleRNN,
Dense, Dropout, concatenate, Layer
from tensorflow.keras.models import Model
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from sklearn.model_selection import train_test_split
from tensorflow.keras.layers import Input, Embedding, SimpleRNN,Bidirectional, Dense,
Dropout, concatenate
from tensorflow.keras.models import Model
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.utils.class_weight import compute_class_weight
from nrclex import NRCLex
from textblob import TextBlob
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from tensorflow.keras.layers import Layer
import tensorflow as tf
from tensorflow.keras.layers import Input, Embedding, Bidirectional, SimpleRNN, Dense,
Dropout, concatenate
from tensorflow.keras.models import Model
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from nltk import pos_tag
from nltk.corpus import wordnet
import nlpaug.augmenter.word as naw
aug = naw.SynonymAug(aug_src='wordnet')
from sklearn.metrics import precision_score, recall_score, f1_score

# NLTK downloads
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('punkt_tab')
nltk.download('brown')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('conll2000')
nltk.download('movie_reviews')
```

```
# Download missing textblob corpora (redundant after specific nltk downloads, but keeping just
in case)
!python -m textblob.download_corpora

# Initialize stop words and lemmatizer
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
# STEP 1: Text cleaning function

# POS tagging conversion
def get_wordnet_pos(treebank_tag):
    if treebank_tag.startswith('J'):
        return wordnet.ADJ
    elif treebank_tag.startswith('V'):
        return wordnet.VERB
    elif treebank_tag.startswith('N'):
        return wordnet.NOUN
    elif treebank_tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN

 def clean_text(text):
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    tokens = text.lower().split()
    tokens = [word for word in tokens if word not in stop_words]
    pos_tags = pos_tag(tokens)
    lemmatized = [lemmatizer.lemmatize(word, get_wordnet_pos(pos)) for word, pos in pos_tags]
    return ' '.join(lemmatized)

# STEP 2: Load dataset
df = pd.read_csv("data.csv")
df.dropna(subset=['reviewText'], inplace=True)
df['reviewText'] = df['reviewText'].astype(str).apply(clean_text)

# STEP 3: Lexicon Features
df['frustration_score'] = df['reviewText'].apply(lambda x:
sum([NRCLex(x).raw_emotion_scores.get(e, 0) for e in ['anger', 'sadness', 'fear']]))
df['polarity'] = df['reviewText'].apply(lambda x: TextBlob(x).sentiment.polarity)
df['subjectivity'] = df['reviewText'].apply(lambda x: TextBlob(x).sentiment.subjectivity)
df['emotion_diversity'] = df['reviewText'].apply(lambda x:
len(NRCLex(x).raw_emotion_scores))
df['label'] = df['frustration_score'].apply(lambda x: 1 if x >= 1 else 0) # Move label creation here

# Step 4: data augmentation
df_aug = df[df.label == 1].copy()
```

```
df_aug['reviewText']=df_aug['reviewText'].apply(lambda x: aug.augment(x)[0] if
aug.augment(x) else x)
df_balanced = pd.concat([df, df_aug])

# STEP 4: Tokenization
max_words = 10000
max_len = 100
tokenizer = Tokenizer(num_words=max_words, oov_token='<OOV>')
tokenizer.fit_on_texts(df_balanced['reviewText'])
sequences = tokenizer.texts_to_sequences(df_balanced['reviewText'])
padded = pad_sequences(sequences, maxlen=max_len)
df_balanced['padded'] = padded.tolist()

extra_features = np.vstack((
  df_balanced['frustration_score'].values,
    df_balanced['polarity'].values,
    df_balanced['subjectivity'].values,
    df_balanced['emotion_diversity'].values
)).T

# STEP 4: Tokenization
from tensorflow.keras.preprocessing.text import Tokenizer

max_words = 10000
max_len = 100
tokenizer = Tokenizer(num_words=max_words, oov_token='<OOV>')
tokenizer.fit_on_texts(df_balanced['reviewText'])
sequences = tokenizer.texts_to_sequences(df_balanced['reviewText'])
padded = pad_sequences(sequences, maxlen=max_len)
df_balanced['padded'] = padded.tolist()

# STEP 5: GloVe Embedding
embedding_dim = 300
embedding_index = {}
with open("glove.6B.300d.txt", encoding="utf8") as f:
    for line in f:
        values = line.split()
        word = values[0]
        coefs = np.asarray(values[1:], dtype='float32')
        embedding_index[word] = coefs

embedding_matrix = np.zeros((max_words, embedding_dim))
for word, i in tokenizer.word_index.items():
    if i < max_words and word in embedding_index:
        embedding_matrix[i] = embedding_index[word]
# Print some info to verify
```

```
print("Total words in GloVe embeddings:", len(embedding_index))
print("Shape of embedding matrix:", embedding_matrix.shape)
df_embedded_matrix.head()

# STEP 6: Split data
X = padded
y = df_balanced['label'].values # Use label from the balanced dataframe

X_train, X_test, X_extra_train, X_extra_test, y_train, y_test = train_test_split(
    X, extra_features, y, test_size=0.2, random_state=42
)

class_weights = dict(enumerate(compute_class_weight(class_weight='balanced',
classes=np.unique(y_train), y=y_train)))

# STEP 7: Attention Layer
class AttentionLayer(Layer):
    def build(self, input_shape):
        self.W = self.add_weight(name="att_weight", shape=(input_shape[-1], 1),
                                 initializer="normal", trainable=True)
        self.b = self.add_weight(name="att_bias", shape=(input_shape[1], 1),
                                 initializer="zeros", trainable=True)
        super().build(input_shape)
def call(self, x):
        e = tf.keras.backend.tanh(tf.keras.backend.dot(x, self.W) + self.b)
        a = tf.keras.backend.softmax(e, axis=1)
        output = x * a
        return tf.keras.backend.sum(output, axis=1)

# Define Input layers
text_input = Input(shape=(max_len,), name='text_input')
frust_input = Input(shape=(extra_features.shape[1],), name='frust_input')

# Embedding + Bi-RNN + Attention
x = Embedding(input_dim=max_words, output_dim=embedding_dim,
              weights=[embedding_matrix], input_length=max_len,
              trainable=True)(text_input)
x = Bidirectional(SimpleRNN(128, return_sequences=True))(x)
x = tf.keras.layers.LayerNormalization()(x)

x = AttentionLayer()(x)
x = Dropout(0.4)(x)

# Combine with lexicon features
concat = concatenate([x, frust_input])
dense1 = Dense(64, activation='relu')(concat)
```

```
drop2 = Dropout(0.3)(dense1)
output = Dense(1, activation='sigmoid')(drop2)

# Final model
model = Model(inputs=[text_input, frust_input], outputs=output)
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
model.summary()
early_stop = EarlyStopping(monitor='val_loss', patience=10, restore_best_weights=True)
reduce_lr = ReduceLROnPlateau(monitor='val_loss', patience=5, factor=0.3, min_lr=1e-6)

# Step
history = model.fit([X_train, X_extra_train], y_train,
                    validation_data=([X_test, X_extra_test], y_test),
                    epochs=100, batch_size=64,
                    class_weight=class_weights,
                    callbacks=[early_stop,reduce_lr],
                    verbose=2)

loss, acc = model.evaluate([X_test, X_extra_test], y_test)
print(f"\nTest Accuracy: {acc:.4f}")

y_pred = model.predict([X_test, X_extra_test])
y_pred_class = (y_pred > 0.5).astype("int32")

# Accuracy plot
plt.plot(history.history['accuracy'], label='Train Acc')
plt.plot(history.history['val_accuracy'], label='Val Acc')
plt.legend(); plt.title('Accuracy per Epoch'); plt.grid(); plt.show()
cm = confusion_matrix(y_test, y_pred_class)
ConfusionMatrixDisplay(cm, display_labels=['Not Frustrated', 'Frustrated']).plot(cmap='Blues')
plt.title("Confusion Matrix")
plt.show()

from sklearn.metrics import precision_score, recall_score, f1_score

# STEP: Calculate Metrics
precision = precision_score(y_test, y_pred_class)
recall = recall_score(y_test, y_pred_class)
f1 = f1_score(y_test, y_pred_class)
accuracy = acc  # Model's test accuracy

# STEP: Plot All Metrics
metrics = [accuracy, precision, recall, f1]
metric_names = ['Accuracy', 'Precision', 'Recall', 'F1-Score']

plt.figure(figsize=(8, 5))
```

```
bars = plt.bar(metric_names, metrics, color=['skyblue', 'lightgreen', 'salmon', 'goldenrod'])
plt.ylim(0, 1.1)

# Annotate values
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2.0, height, f'{height:.2f}', ha='center', va='bottom',
fontsize=12)

plt.title("Model Evaluation Metrics")
plt.ylabel("Score")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# STEP: Print Metrics as a Table
metrics_dict = {
    "Metric": ["Accuracy", "Precision", "Recall", "F1-score"],
    "Value": [accuracy, precision, recall, f1]
}

df = pd.DataFrame(metrics_dict)
print(df.to_string(index=False))

# STEP 11: Evaluation
y_pred = (model.predict([X_test, X_extra_test]) > 0.5).astype(int)
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=["Not Frustrated", "Frustrated"]
```